STRUCTURAL DYNAMICS RESEARCH

Traffic Organization and Bridge Deck Response

Vehicle–bridge interaction, spatial sampling and stress-cycle screening

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Research question

How does traffic organization change deck response when the vehicle fleet stays fixed, and how reliably do saved locations capture local cycle demand?

Computational model

26,520

MITC4 bending/shear plate elements

Coupled steel beams and vertical supports
Approximately 79,600 algebraic coordinates
Linear equivalent model in MATLAB

Study design

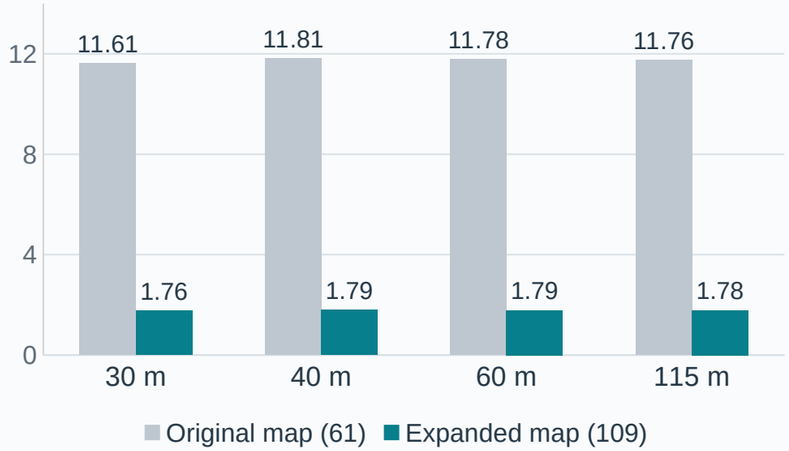
22 selected traffic scenarios use the same 240-vehicle fleet, including 20 heavy vehicles.

The comparison varies longitudinal spacing, lateral spread and vehicle ordering, with two-lane and grouped-traffic references.

Spatial sampling sensitivity

Five cases share an expanded recording map: 61 to 109 interior locations. Additional-output reruns reproduce every original moment history with zero difference.

Gross-section nominal stress: maximum I_3 ratio



Four uniform-order cases at lateral spread $S_Y = 0.05$. Each ratio uses its own recording map for both the case and reference. Maxima can occur at different locations.

1.76–1.79

Expanded-map I_3 ratios replace the original 11.6–11.8 values. Coverage of the reference matters.

Numerical evidence

20 initial cases passed completeness and contact-convergence audits. Iterations remained within the declared tolerance.

Five recording-only reruns preserve all 213 original scalar histories.

Section recovery extension

119 saved coordinates now have nominal reinforcement-section mappings.

952 section variants and 3,808 independent unit-resultant checks give a maximum residual of 1.1×10^{-14}.

Interpretation and next steps

$I_3 = \Sigma\, n_i\, (\Delta\sigma_i / 1\ \mathrm{MPa})^3$
This weights rainflow stress ranges for screening.

Local convergence, section assumptions and material fatigue calibration limit physical conclusions. Unrecorded point histories remain unavailable.

Individual work: model reconstruction, implementation, verification, simulation review and documentation. Professor Zhang supplied the bridge drawings.

Preliminary numerical case study. Private framework and source drawings excluded. Experimental validation and absolute fatigue life remain open.